PERFORMANCE TEST ON PELTON WHEEL

AIM:-

To conduct the experiment on Pelton wheel and determine the efficiency of the turbine and plot the following graphs.

1. Output v/s Discharge
2. Input v/s Discharge.
3. Head v/s Discharge
4. Efficiency v/s Discharge.
5. Efficiency v/s Brake power (at 5 kg load)

APPARATUS

1. Pelton wheel turbine
2. Venturimeter
3. Centrifugal pump to supply water to the turbine.
4. Brake drum with loading arrangement.
5. Meter scale.

PRINCIPLE

$$\text{Efficiency} = \frac{Output}{Input} \times 100\ \%$$

$$\text{Output power} = \frac{2\pi N (w-s)\, g\, R}{60{,}000}\ \text{kW}$$

Where

N = Speed of the turbine in rpm.
W = load applied in Kg.
S = Slacking load in Kg.
R = Radius of the brake drum + Radius of the rope in m.

$$\text{Input power,}\quad IP = \frac{\rho\, g\, Q\, H}{1000}\ \text{kW}$$

Where

ρ = Density of water = 1000 $\frac{kg}{m^3}$
g = acceleration due to gravity = 9.81 $\frac{m}{s^2}$
H = Turbine head in m of water.
Q = Discharge of water (m^3/s)

OBSERVATIONS

Sl. No.	Turbine Pressure	Turbine Head	h_1	h_2	h	W	S	W-S	Speed	Q	BP	IP	Efficiency
	kg/cm^2	m	cm	cm	m	kg	kg	kg	rpm	m^3/s	kW	kW	%
1													
2													
3													
4													
5													
6													

SAMPLE CALCULATION (Set No:. ...)

Brake power, BP	$= \frac{2\pi N (w-s) Rg}{60{,}000}$	kW.
Speed of Turbine, N	= rpm	
Effective load, $(w-s)$	= (....-.....)=	kg
Radius of Brake drum, R_1	=	m
Radius of rope, R_2	=	m
Effective radius, R	$=(R_1+R_2)$	m
Effective radius, R	=	m
Acceleration due to gravity, g	= 9.81	m/s^2
Brake power, BP	=	kW
Input Power, IP	$= \frac{\rho g Q H}{1000}$	kW
Density of water, ρ	= 1000	kg/m^3
Acceleration due to gravity, g	= 9.81	m/s^2
Discharge ,Q	$= Cd\ K\sqrt{2gh}$	m^3/s
Coefficient of discharge, Cd	= 0.93	
Constant, K	$= \frac{a_1 a_2}{\sqrt{a1^2 - a2^2}}$	
Diameter of inlet, d_1	=	m
Area of inlet, a_1	=	m^2
Diameter of throat, d_2	=	m
Area of throat, a_2	=	m^2
Constant, K	=	
Manometer difference, h	$= (h_2 - h_1)\frac{(S_m - S_l)}{Sl}$	
Left limb manometer reading, h_1	=	cm
Right limb manometer reading, h_2	=	cm
Manometer difference, h	=	m of water
Discharge, Q	=	m^3/s
Turbine head, H	=	kg/cm^2 = ... m of water
Input power, IP	=	kW
Efficiency of turbine, η	$= \frac{Brake\ power}{Input\ power}$ 100% =.......	%

Discharge of water, $Q = Cd \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}} \sqrt{2gh} \ \frac{m^3}{s}$

Where

a_1 = Area of the inlet of the venturimeter in m^2

a_2 = Area of throat of venturimeter in m^2

g = Acceleration due to gravity in m sec^2

h = manometer difference in terms of Hg

= $(h_2 - h_1) \times 12.6$ m of water

h_1 = Left limb manometer reading in m

h_2 = Right limb manometer readings in m

PROCEDURE

1. Start the pump and allow the water to flow through the Turbine from the sump.
2. Take the readings at no load.
3. Make the speed constant at 1300 rpm by regulating the inlet valve. (Measure the rpm using a tachometer).
4. Note the brake load, load on the brake drum W and Spring balance reading S
5. Note the inlet (Turbine) pressure. (Turbine Head)
6. Note the differential manometer readings (h_1& h_2)
7. Repeat the above procedure by increasing the load on brake drum at constant speed at 1300 rpm.(say 2 kg. 4, 6, 8, 10)
8. Tabulate the observations and calculate the efficiency of the turbine.

GRAPHS

1. Output v/s Discharge
2. Input v/s Discharge.
3. Head v/s Discharge
4. Efficiency v/s Discharge.
5. Efficiency v/s Brake power (at 5 kg load)

RESULT

The experiment on the Pelton wheel was conducted. The efficiency of the pelton wheel is determined and the required graphs were plotted.

INFERENCE